

Research Question

Can editing audio distort someone's voice enough to make it effectively anonymous, and how do we model the trade-off between privacy and intelligibility in anonymized speech data?

Brief overview:

Nowadays, most of us have voice assistants. Whether it's Siri and/or Alexa (fitting...), our voices are used daily to administer everyday commands. How much protection is put into making our voices anonymous before being processed by our smart devices? In this experiment, we aim to see how many levels of abstraction it takes before our voices are unrecognizable. Will the trade-offs be worth it, how much can we manipulate audio before it is incomprehensible, and is anonymity truly achievable when related to audio?

High-level criteria

1. Technical depth:

We plan to use Python audio libraries like librosa, pydub, and scipy to manipulate and analyze our recordings. Like in our last project, we will be collecting data that will consist of 5-10 short audio clips from as many people as we can force into participating. We will mathematically model each voice by extracting features like pitch, timbre, and rhythm. Then, we'll manipulate the audio by tweaking features, speed, compressing, adding noise, etc. After doing this, we will measure how these anonymization methods change the distance between audio vectors. We will then assess the trade-offs and try to find the balance of audio that can still be comprehended and understood while having anonymized samples. We will use a cosine similarity comparison to see how far we can get from 1 with the altered audio vs the original samples.

Challenges:

Once again our project will be dependent on the data we collect, which presented as a problem last time. However these audio clips should be more manageable than our last project, especially given the fact that we have almost a month including Thanksgiving break. Another potential challenge is that neither of us have worked with manipulating audio before, but we have already found a couple libraries we previously mentioned like scipy and librosa to make that easier.

2. Topic relevance:

In lecture, we talked about biometrics (or "something you are") as one of the categories of information that can be used to authenticate someone. Physical traits like fingerprints, or in this project voice, are some of the most intuitive and oldest ways to prove identity. But they also raise the biggest privacy concerns. Making authorized transactions or

proving your identity in other ways with voice assistants is convenient, but one of their biggest complaints is how extremely creepy and intrusive they are. More importantly, once biometrics are given they can't be revoked, and unless you burn off your fingertips or get plastic surgery they cannot be changed either. In our research we found that Amazon and Apple can keep user recordings from their voice assistants for up to two years, and in Amazon's case sometimes indefinitely. Even when companies try to anonymize voice data by associating it with a random identifier instead of a specific account, the unaltered recordings themselves still contain sensitive and identifiable information about their users. This could be misused and used to impersonate or re-identify someone unintentionally and without their consent.

3. Support of goals:

Alexa:

- **Personal Agency:** For my goal of personal agency, this topic feels weirdly personal to me. I share a name with one of the most popular voice assistants, and I'm constantly reminded of that fact every time I meet someone new. Almost every time someone finishes their initial joke, I'm asked my opinion on voice assistants as if I have some credibility. So throughout the past few years, it has naturally guided my interest. I think having something listening in at all times, even just for a trigger word, in your home is insanely creepy. Enough so that I unplugged my family's home Alexa (or Echo, since we had to change its nickname), which nobody has even noticed for years. Even though I have a special hatred in my heart for Amazon's Alexa, only when I began this project did I realize how hypocritical I was by constantly using Siri. Arguably, my phone is with me more and is more invasive than home assistants, but I ignore that fact in favor of convenience when I'm too lazy to write a text. I hope this project will help me either gain some peace of mind on how my voice data is being used, or at least gain the agency to decide with confidence whether changing the song while driving is worth having Siri listen to me in the same way I look down upon my family when using Alexa.
- **Research Engagement:** Like I said in our last proposal, research engagement very obviously fits as a goal for a research project like this. Now both Gaby and I have at least a little experience with planning, pivoting, and completing research. Still, not one project is ever the same and comes with its own set of drawbacks.
- **Career Applications:** For our last project, we did not have to do as much compromising because we landed on our idea quickly by chance and both were excited. Though we definitely benefit from group projects by knowing each other so well, we had a harder time settling on one research question for this unit and had to exercise skills like compromise. I think we also have to opportunity to able to learn from some of the mistakes we made before like time management on collecting our data.

Gaby:

- **Career application** - Once again, we are stepping out of our comfort zone and tackling an issue that we had no idea how it existed. Researching for this project has helped our communication skills and critical thinking. Something that I love about taking this class

with Alexa is that every time we have an assignment or project it always sparks so much conversation between us and we always land on such fun ideas that we are both interested in. Being able to effectively communicate and compromise like this is something that I think is going to be extremely applicable to our careers.

- **Research engagement** - To be honest I'm struggling to not sound redundant. This project has definitely taught me a lot already that is very applicable to the real world even though we are just beginning. We started from scratch with zero ideas and made both of our points of views work into something we are both interested in. We both keep constantly growing during these projects/ assignments and are gaining invaluable experience with open ended deliverables.
- **Privacy competency** - Once again we picked a project about something that we really didn't know much about. We have already learned so much about the steps that go into protecting our voices and I expect to learn so much more. This project should particularly help hone our python skills since that is what we will be coding in for the majority of this project.

Outline of Write-up

W0:

We will introduce our research question and a general outline of our experiment. Our problem is to see how anonymous voice samples can be before we lose the meaning of the audio, and what the trade-off is when it comes to audio being private and usable. The expected input of our experiment is audio samples from our friends saying simple commands like "turn on the lights" or "play music". Our expected output will be whether or not a speech recognition API can recognize the altered data and the results of our cosine similarity comparison.

W1:

We will examine research on existing privacy procedures for popular voice assistants in order to establish the importance and "why" of our experiment. After doing some preliminary research, specifically on Amazon Alexa protocols, we couldn't find the specific ways that our audio is being recorded. Most Amazon websites say that yes, your voice is being recorded and uploaded to AWS, and that it is also used as a way to improve accuracy. This text usually also comes with a way of deleting stored voice recordings... So is privacy not something that people are worried about anymore? Our voice recordings in raw, unedited form is something that can be very powerful and usually can not be drastically changed. Voice is a very identifiable factor of ourselves and we should know if it can be protected and how it is being stored.

W2:

We will talk about how we collected our participants and audio data to be analyzed. In our last project, data was a huge bottleneck for us. Since we have a more generous time frame and will both be going home for the holidays (and therefore have family to force

into participating), we expect to be able to solve this problem. We plan to have at least 10 participants, and for each person to provide multiple clips consisting of the same short commands repeated at least 5 times. These commands will be similar to what voice assistants hear, like “what’s the weather?” or “turn on the lights”. Repeating the same command multiple times per participant will help minimize random fluctuations and let us average the features of their voice.

C3:

In this part, we will talk about the exact features we used and develop the code that will extract those from our audio clips. As mentioned before, we know of a couple Python libraries that might be useful to make the samples as anonymous as we can without compromising the intelligibility.

C4:

In this section, we will show and explain the code we made to anonymize the voice samples through multiple methods. We plan to use compression, changing pitches, adding background noise, and changing the speed of the audio. We expect to use different combinations of all of these to get better/worse results.

W5:

In this section we will evaluate the results from the altered audio. We want to see which method has the best result. We will measure “best” by two things, we want to see how anonymized the audio is and how understandable it is. In order to analyze whether the speech is unintelligible using speech recognition API (like Whisper or Google Speech). We will measure how anonymized the audio is to use a cosine similarity comparison. We will compare the vectors of the original voice clips with the anonymized data and see how far from 1 we can get.

W6:

Some shortcomings that we can see happening is the audio clips still being recognizable after being altered. We are worried that changing the clips will make the audio incomprehensible, thus making it easier to re-identify the audio. We know that even in the worst case the audio will still be altered but we are still worried that with advanced technology (like what Amazon, Google, or Apple would use) our voices will still be identifiable. Some pivots we are willing to take are looking at encryption for audio instead. The issue with that is, unlike anonymizing audio, encrypted audio is useless unless it's decrypted. So at that point, yes our audio is protected, but to make it useful to a company they would need to decrypt it.